Spring AI: Beginner to Guru

Generative AI with Spring

Introduction to Large Language Models

Introduction to Large Language Models

- Introduction to LLMs
  - LLMs are neural networks trained on massive text data to understand/generate natural language
    - Data gathered from scraping websites, articles, books - just about any source of text
  - Different from traditional programming which gives explicit instructions
  - LLMs learn patterns from data to solve problems that rules-based coding cannot

Introduction to Large Language Models

- How LLMs Work
  - Process broken into 3 steps:
    - Tokenization - splitting text into tokens (pieces of words)
    - Embeddings - converting tokens to numerical vectors
    - Transformers - using algorithms like multi-head attention to process vectors
  - Use vector databases to store/retrieve word embeddings and relationships
  - Predict next word based on context from training data

Introduction to Large Language Models

- LLM History and Evolution
  - Started with simple language models like ELIZA in 1966
  - Transformers in 2017 enabled training of large models like GPT, BERT
  - Rapidly increased in scale (GPT-3 175B parameters in 2020)
  - Recent models like GPT-4 use mixture of experts, multimodality

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- Training LLMs
  - Requires massive datasets of text from web pages, books, etc.
  - Data preprocessing to clean, transform training data
  - Computationally expensive training process on specialized hardware
  - Tuning based on metrics like perplexity, human feedback

Introduction to Large Language Models

- Fine-tuning
  - Further training large foundational models for specific use cases
  - E.g. fine-tune on pizza order conversations for pizza chatbot
  - Allows customization while leveraging pre-trained capabilities

Introduction to Large Language Models

- Limitations and Challenges
  - Bias, hallucinations, mathematical reasoning weaknesses
  - Limited knowledge cutoff, mitigated by web browsing
  - High computational cost, potential copyright violations
  - Potential misuse for misinformation, disrupting workforce

Introduction to Large Language Models

- Applications
  - Natural language tasks like translation, summarization, Q&A
  - Programming assistance, creative writing
  - Essentially any task done on a computer

Introduction to Large Language Models

- Current Research
  - Knowledge distillation to smaller efficient models
  - Retrieval augmented generation to query external data
  - Multimodality with voice, image, video inputs
  - Improving reasoning ability with step-by-step thinking
  - Increasing context window size with external memory

Introduction to Large Language Models

- Ethical Considerations
  - Copyright issues from training data
  - Potential harmful uses like scams, disinformation
  - Need for AI alignment with human values as systems become more capable

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